COLLEGE BUSINESS EDUCATION

DAR ES SALAAM

GROUP ASSIGNMENT

COURSE: BACHELOR DEGREE IN INFORMATION TECHNOLOGY.

MODULE: Database Management and Administration

MODULE CODE: ITU07307.

FACILITATOR: Kikwembo Mlekwa.

YEAR OF STUDY: Second Year.

SEMESTER : 1st

ACADEMIC YEAR: 2025/2026.

STREAM: TNC.

GROUP NUMBER EIGHT.

GROUP MEMBERS

SN	FULL NAME	REGISTRATION NUMBER
1	FATMA MOHAMED MUSA	03.9607.02.01.2024
2	FRANCE GAUFRID FUSSY	03.3154.01.01.2024
3	GOODLUCK JULIUS ABRAHAM	03.1535.02.01.2024
4	JOSHUA M SAMWEL	03.4813.02.01.2024
5	REGINA CHARLES	03.4512.02.01.2024
6	VICTORIA EMANUEL MBOKO	03.3559.01.01.2022
7	WILSON MAPE MAPANDE	03.4228.02.01.2024

AIRPORT DEPARTMENTS AND THERE OPERATIONS

Department/ Agency	Base	Duties	Objectives
1. CUSTOMS	(a) At departure Lounge:	• Checks on illegal exportation of goods • Checks on illegal exportation of Currency • Checks on illegal exportation of Drugs • Checks on illegal exportation of Arms	• To protect revenue, lev,y etc., compile statistics. • To protect territorial integrity (economy) • To protect the security of the citizens of destination. • To protect revenue (customs duty and sales tax) • To compile statistics – trade trends.
	(b) At Arrival Lounge:	• Checks on illegal importation of goods	• To protect the national morals by seizing pornographic materials. • To protect the national healthy by seizing dangerous drugs. • To protect national security by seizing illegal importation of arms.

	(c) At Cargo Terminal	Check on the legal importation /exportation of goods.	• To raise revenue – as Customs & Sales Tax • To compile statistical data – trade trends.
2. IMMIGRATION	(a) At Departure /Arrival lounge	• Check on the movement of passengers	• To protect national integrity/Security by ensuring that passengers are authorized entry/ exit at origination/destinations.
3. HEALTH	(b) Arrival Lounge	• Detain and quarantine passengers without valid medical/health certificates; and • Inoculates or treats them.	• To protect the national health of the host country • To protect the health of the nation by ensuring that health formalities of Imports/exports are complied with.
4. CIVIL AVIATION	• At Control Tower • Lays down rules of operators, crew, aircraft airworthiness, aerodrome worthiness etc.)	• Controls the movements of aircraft in the skies	• Security of aircraft and occupants assured • Protects the national integrity of National airspace

5. METEOROLOGICAL DEPARTMENT	• At Airport	• Briefs operating crew on the weather conditions of its flight path	• Security and safety of aircraft passengers
6. FIRE BRIGADE	• At Airport	• Maintain fire services	• Safety of aircraft and occupants.
7. AIRPORT AUTHORITY	• At Airport	• Maintain airports to serviceable Standards as directory by the Civil Aviation. • Maintain statistics of the traveling public.	• Safety of aircraft and occupants. • Determine long and short time protectorates for expansions and improvements of aircraft for the safety of passengers.
8. POLICE/ MILITARY	At Airport	• Maintain law and order	• Protect territorial integrity • Protect and ensure security of airport users and aircraft.
9. BANKS/ BUREAU DE CHANGE	At Airport	• Money exchangers	• Protect territorial integrity by maintaining proper and legal money exchanges.

DATA MODELING

From the Airport Operations The following are the important data used to create a database

Entinty: Airlines

- Airline_Id(PK)INT
- Name(VARCHAR)
- Registration_number(VARCHAR)

Entinty: Aircraft
Attributes:

- Aircraft_Id(PK)INT
- Registration number(VARCHAR)
- Aircraft_type(VARCHAR)
- Airline_Id(INT)Fk REFERENCES Airlines(Airline_Id)

Entinty: FlightInformation

Attributes:

- Flight_Id(PK)INT
- Flight number(VARCHAR)
- Depature_Airport(VARCHAR)
- Destination_Airport(VARCHAR)
- Route(VARCHAR)
- Departure_time(Datetime)
- Arrival_Time(Datetime)
- AircraftId(INT)FK REFERENCES Aircrafts(Aircraft_Id)

Entinty: Passangers

Attributes:

- Passenger_Id(PK)INT
- Name(VARCHAR)
- Address(VARCHAR)
- Email(VARCHAR)
- Phone(VARCHAR)
- Gender(VARCHAR)
- Nationality(VARCHAR)
- National_ID(VARCHAR)

Entinty: Tickets

Attributes:

- Ticket_Id(PK)INT
- Flight_Id(INT)FK REFERENCES FlightInformation(Flight_Id)
- Passenger_Id(INT)FK REFERENCES Passengers(Passenger_Id)
- SeatNumber(VARCHAR)
- TravelType(VARCHAR)

Entinty: ImmigrationRecords

Attributes:

- Migration_Id(PK)INT
- Ticket_Id(INT)FK REFERENCES Tickets(Ticket_Id)
- Passenger_Id(INT)FK REFERENCES Passengers(Passenger_Id)
- PassportNumber(VARCHAR)
- VisaNumber(VARCHAR)

- Type(VARCHAR)
- ClearanceDate(DateTime)

Entinty: Luggages

Attributes:

- Luggage_Id(PK)INT
- Ticket_Id(INT)FK REFERENCES Tickets(Ticket_Id)
- Weight(INT)
- Luggage_type(VARCHAR)
- Status(VARCHAR)
- Date(DateTime)

Entinty: Cargos

Attributes:

- Cargo_Id(PK)INT
- Flight_Id(INT)FK REFERENCES FlightInformation(Flight_Id)
- SenderName(VARCHAR)
- ReciverName(VARCHAR)
- CargoWeight(INT)
- CargoType(VARCHAR)
- Date(DateTime)

Entinty: Inspections

Attributes:

- Inspection_Id(PK)INT

- Luggage_Id(INT, nullable)FK REFERENCES Luggages(Luggage_Id)
- Cargo_Id(INT, nullable)FK REFERENCES Cargos(Cargo_Id)
- Inspection_Type(VARCHAR)
- InspectionDate(Date)
- Status(VARCHAR)

Entinty: Departments

Attributes:

- Department_Id(PK)INT
- Name(VARCHAR)

Entinty: Positions

Attributes:

- Position_Id(PK)INT
- Name(VARCHAR)

Entinty: Roles

Attributes:

- Roles_Id(PK)INT
- Name(VARCHAR)

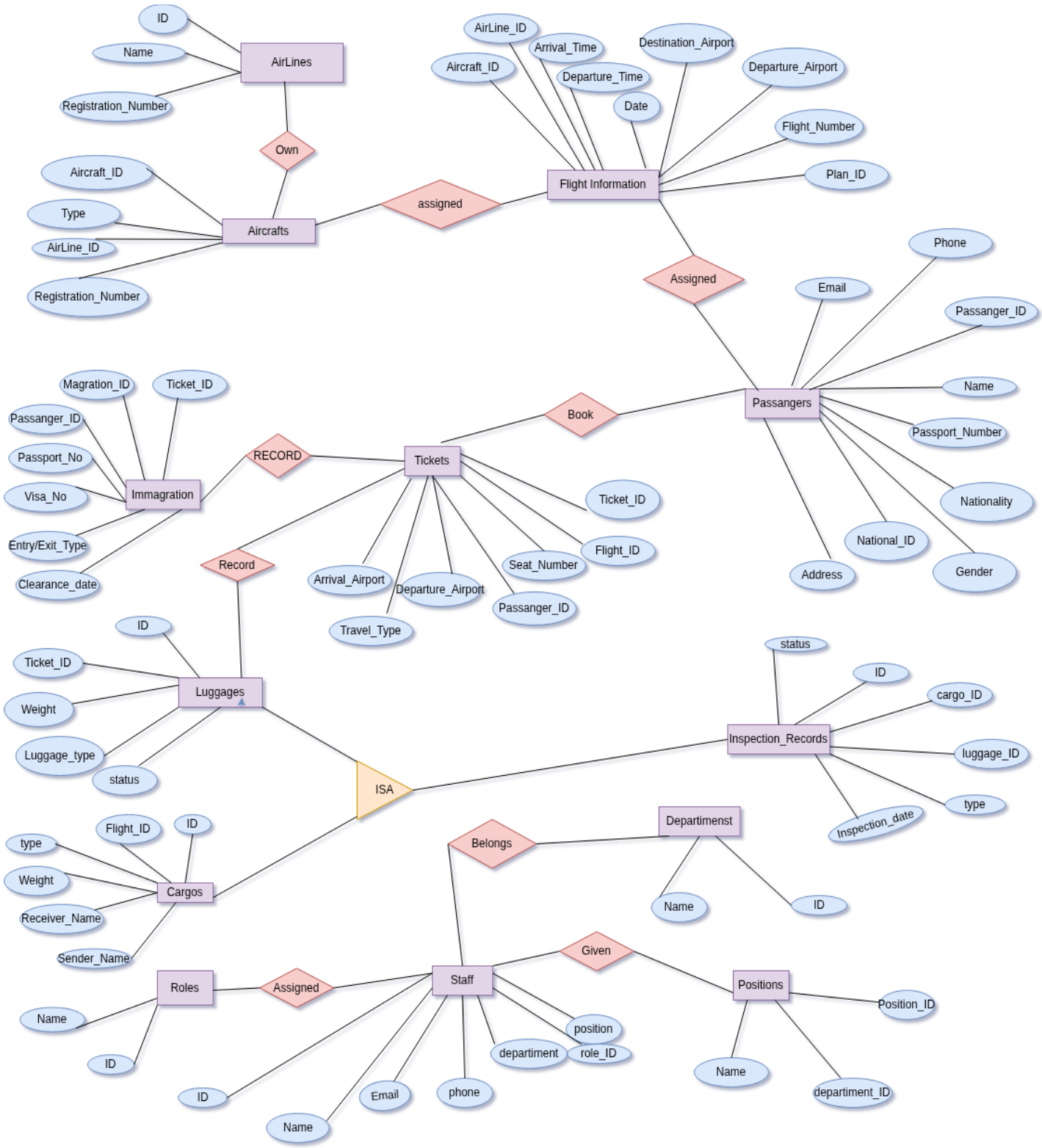
Entinty: Staffs

Attributes:

- Staff_Id(PK)INT
- Name(VARCHAR)

- Email(VARCHAR)
- Phone(VARCHAR)
- Department_Id(INT)FK REFERENCES Departments(Department_Id)
- Role_Id(INT)FK REFERENCES Roles(Role_Id)
- Position_Id(INT)FK REFERENCES Positions(Positions_Id)

ER Diagrams


AirLines
ID
Name
Registration_Number
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Aircrafts
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Type
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Registration_Number
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Flight Information
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Arrival_Time
Departure_Time
Destination_Airport
Date
Departure_Airport
Flight_Number
Plan_ID
Assigned
Passangers
Phone
Email
Passanger_ID
Name
Passport_Number
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Arrival_Airport
Travel_Type
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Passanger_ID
Passport_No
Visa_No
Entry/Exit_Type
Clearance_date
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Ticket_ID
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Luggage_type
status
ISA
Inspection_Records
status
ID
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Weight
Receiver_Name
Sender_Name
Belongs
Departimenst
Name
ID
Given
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Roles
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Positions
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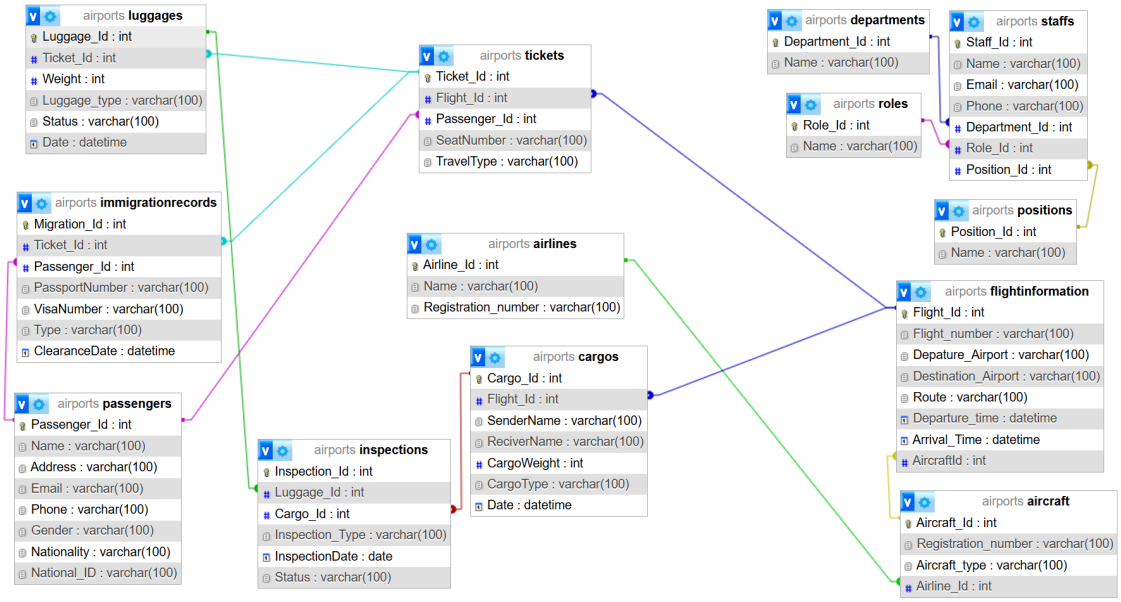
departiment_ID

DATABASE SCHEMA